

The Why and How of Stochastic Gradient Descent

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Summary

1 The “Why” of Stochastic Gradient Descent

- Background
- Origins and Benefits of Stochasticity

2 Problem Formulations

- Classes of Objective Functions
- Quick Primer on Stochastic Processes

3 Convergence Theorems

- Algorithms Studied
- Convergence Theorems for Unified Algorithm
- Numerical Results and Future Research Topics

Outline

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Problem Formulation

We wish to minimize a $\mathcal{C}^1$ function $J : \mathbb{R}^d \rightarrow \mathbb{R}$. Assumptions on $J(\cdot)$

- $J(\cdot)$ is bounded below.
 - This rules out functions like $J(\theta) = \langle \mathbf{y}, \theta \rangle$.
 - Define

$$J^* = \inf_{\theta} J(\theta).$$

By replacing $J(\cdot)$ by $J(\cdot) - J^*$, it can be assumed that $J^* = 0$.

- The infimum is attained. This rules out functions like $J(\theta) = e^{-\theta}$.



Problem Reformulation

- Observe that if θ^* is a global minimizer,¹ then $\nabla J(\theta^*) = \mathbf{0}$. But the converse is not always true.
- In this talk the focus is on solving $\nabla J(\theta^*) = \mathbf{0}$.
- Are there classes of functions (other than convex functions) where every stationary point θ^* such that $\nabla J(\theta^*) = \mathbf{0}$ is also a global minimizer?
- Yes, and more on that later.

¹My convention: If $J(\theta^*) = J^*$, then J^* is the global *minimum*, while θ^* is the global *minimizer*.

Classical Optimization: Steepest Descent

Classical optimization methods began in the 1960s with the availability of digital computers and programming languages.

Steepest Descent method is centuries old, and predates the advent of computers.

- Start with an initial guess θ_0 .
- At iteration $t + 1$, update θ_t via

$$\theta_{t+1} = \theta_t - \alpha_t \nabla J(\theta_t),$$

where α_t is the “step size” or the “learning rate,” which can be a function of the iteration counter t .

- Repeat.



Classical Optimization: Momentum-Based Methods

- Even in the 1960s, it was obvious that Steepest Descent converged very slowly.
- By the 1980s, Steepest Descent was “no longer recommended.”
- This led to the advent of conjugate gradient, quasi-Newton, and momentum-based methods.
- For conjugate gradient and quasi-Newton methods, see references on the next slide.
- After that, we introduce momentum-based methods.

Some References

- J. S. Kowalik and M. R. Osborne, *Methods for Unconstrained Optimization Problems*, Elsevier, 1968.
- Roger Fletcher, *Practical Methods of Optimization, Volume 1: Unconstrained Optimization*, Wiley, 1980.
- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.
- Yurii Nesterov, *Introductory Lectures on Convex Optimization: A Basic Course*, Springer, 2004

Heavy Ball Method

- Introduced by (late) Boris Polyak in 1964.
- The update rule is

$$\theta_{t+1} = \theta_t - \alpha_t \nabla J(\theta_t) + \mu_t (\theta_t - \theta_{t-1}),$$

where μ_t is the **momentum parameter**.

- This reformulation accelerates convergence for quadratic $J(\cdot)$ and is also applicable to strongly convex $J(\cdot)$.

Nesterov’s Accelerated Gradient Method (NAG)

- Introduced by Nesterov in 1983.
- The update rule is

$$\theta_{t+1} = \theta_t - \alpha_t \nabla J[\theta_t + \mu_t(\theta_t - \theta_{t-1})] + \mu_t(\theta_t - \theta_{t-1}).$$

- With this method, $J(\theta_t) = O(t^{-2})$.
- For any convex $J(\cdot)$, no *linear* method can achieve faster convergence, where a “linear method” is one where the search direction at step t is a linear combination of $\nabla J(\theta_\tau), \tau = 0, \dots, t$.

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Types of Objective Functions

A typical objective function studied today has the form

$$J(\boldsymbol{\theta}) := E_{x \sim \pi}[f(x, \boldsymbol{\theta})] = \int_{\mathcal{X}} f(x, \boldsymbol{\theta}) \pi(dx),$$

where $\mathcal{X}$ is some set, π is some probability measure on $\mathcal{X}$, and $f : \mathcal{X} \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a $\mathcal{C}^1$.

Depending on the size and nature of the set $\mathcal{X}$, stochasticity may be unavoidable, and sometimes even desirable.

Example: Neural Network Training

Neural network architecture: A family of maps $H(\cdot, \theta) : \mathbb{R}^n \rightarrow \mathbb{R}$ for each $\theta \in \mathbb{R}^d$, where θ is the “weight” vector.

Training Inputs: labelled pairs $\{(\mathbf{x}_i, y_i)\}_{i=1}^m$, where $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$.

Objective function:

$$J(\theta) := \frac{1}{m} \sum_{i=1}^m L(y_i, H(\mathbf{x}_i, \theta)),$$

where $L : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_+$ is the “loss function” quantifying the mismatch between the actual output $H(\mathbf{x}_i, \theta)$ and the desired output y_i .

Batch Optimization in the General Case

In general

$$J(\boldsymbol{\theta}) := E_{x \sim \pi}[f(x, \boldsymbol{\theta})] = \int_{\mathcal{X}} f(x, \boldsymbol{\theta}) \pi(dx).$$

To apply gradient descent, we must compute $\nabla J(\boldsymbol{\theta}_t)$. Is it true that

$$\nabla J(\boldsymbol{\theta}) = E_{x \sim \pi}[\nabla_{\boldsymbol{\theta}} f(x, \boldsymbol{\theta})]?$$

Not always!

Batch Optimization in the Additive Case

If

$$J(\boldsymbol{\theta}) := \frac{1}{m} \sum_{i=1}^m L(y_i, H(\mathbf{x}_i, \boldsymbol{\theta})),$$

then

$$\nabla J(\boldsymbol{\theta}) = \frac{1}{m} \sum_{i=1}^m \nabla_{\boldsymbol{\theta}} L(y_i, H(\mathbf{x}_i, \boldsymbol{\theta})).$$

This is known as **batch optimization**.

In LLMs, $m \sim 10^{14}$, so implementing the above can be very costly.

Stochastic Gradient Descent

The updating rule is

$$\theta_{t+1} = \theta_t - \alpha_t \mathbf{h}_{t+1},$$

where $\mathbf{h}_{t+1}$ is a *random* vector called the *stochastic gradient*.

We give some examples of stochastic gradients.

Minibatch Optimization

- At step t , choose an integer N_t (possibly random), and choose N_t samples $x_j \in \mathcal{X}, j \in [N_t]$ *with replacement*.
- Define the search direction $\mathbf{h}_{t+1}$ as

$$\mathbf{h}_{t+1} = \frac{1}{N_t} \sum_{j=1}^{N_t} \nabla_{\boldsymbol{\theta}} f(x_j, \boldsymbol{\theta}_t).$$

Then

$$E_{x_{t+1} \sim \pi}[\mathbf{h}_{t+1}] = E_{x_{t+1} \sim \pi}[\nabla_{\boldsymbol{\theta}} f(x_{t+1}, \boldsymbol{\theta}_t)].$$

This is called **minibatch optimization**. Note that the search direction $\mathbf{h}_{t+1}$ is random.



Zeroth-Order Methods

Also known as “gradient-free” methods, these methods approximate the true gradient $\nabla J(\boldsymbol{\theta}_t)$ *using only two function evaluations*.

For each index $t + 1$, suppose $\Delta_{t+1,i}, i \in [d]$ are d different and pairwise independent **Rademacher variables**.²

²Recall that Rademacher random variables assume values in $\{-1, 1\}$ and are independent of each other.

Zeroth-Order Methods (Cont'd)

Let $\Delta_{t+1} \in \{-1, 1\}^d$ denote the vector of Rademacher variables at time $t + 1$. Then the search direction $\mathbf{h}_{t+1}$ is defined componentwise, via

$$h_{t+1,i} = \frac{[J(\boldsymbol{\theta}_t + c_t \Delta_{t+1}) + \xi_{t+1,i}^+] - [J(\boldsymbol{\theta}_t - c_t \Delta_{t+1}) - \xi_{t+1,i}^-]}{2c_t \Delta_{t+1,i}},$$

where $\xi_{t+1,1}^+, \dots, \xi_{t+1,d}^+, \xi_{t+1,1}^-, \dots, \xi_{t+1,d}^-$ represent the (possible) measurement errors.

Block-Updating

At time $t + 1$, the usual update formula is

$$\theta_{t+1} = \theta_t - \alpha_t \mathbf{h}_{t+1}.$$

It is possible to reduce memory calls by updating *only some, but not all* components of θ_t at step $t + 1$.

Three other types of updates are proposed, using the concept of the **Hadamard product** between two vectors of equal dimensions.³

$$(a \circ b)_i = a_i \cdot b_i.$$

³Option 1 is the full coordinate update.

Single Coordinate Update

At time t , choose an index $\kappa_t \in [d]$ at random with a uniform probability, and independently of previous choices. Let $\bar{e}_{\kappa_t}$ denote the elementary unit vector with a 1 as the κ_t -th component and zeros elsewhere. Then define

$$\mathbf{h}_{t+1}^{(2)} = d\bar{e}_{\kappa_t} \circ \mathbf{h}_{t+1}.$$

Multiple Coordinate Update

At time t , choose N different indices κ_t^n from $[d]$ *with replacement*, with each choice being independent of the rest, and also of past choices. Moreover, each κ_t^n is chosen from $[d]$ with uniform probability. Then define

$$\mathbf{h}_{t+1}^{(3)} := \frac{d}{N} \sum_{n=1}^N \bar{\epsilon}_{\kappa_t^n} \circ \mathbf{h}_{t+1}.$$

Bernoulli Update

At time t , let $\{B_{t,i}, i \in [d]\}$ be independent Bernoulli processes with success rate ρ_t . Thus

$$\Pr\{B_{t,i} = 1\} = \rho_t, \forall i \in [d].$$

Then define

$$\begin{aligned} \mathbf{v}_t &:= \sum_{i=1}^d \bar{\epsilon}_i I_{\{B_{t,i}=1\}} \in \{0, 1\}^d, \\ \mathbf{h}_{t+1}^{(4)} &= \frac{1}{\rho_t} \mathbf{v}_t \circ \mathbf{h}_{t+1}. \end{aligned} \tag{1}$$

Summary

- Stochasticity can arise due to noisy measurements of the function and/or gradient.
- In large-scale optimization problems, the *deliberate* introduction of stochasticity can actually *reduce* the computational and/or memory burden at each iteration.
- We have presented three different methods for doing this: minibatch, gradient-free, and block-updating.
- These methods can be used in *any* combination.
- Each of them leads to the search direction $\mathbf{h}_{t+1}$ being a random vector.
- Consequently, algorithms result in *$\mathbb{R}^d$ -valued stochastic processes*, not sequences of vectors in $\mathbb{R}^d$.
- This makes the analysis more tricky.

This concludes the “why” of SGD. Next we study the “how.”



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Details About Book and Course

- In August 2024, I offered (for the third time) a course with the title "Stochastic Algorithms for Nonconvex Optimization and Reinforcement Learning" at IITH:
- The URL for the course is
https://people.iith.ac.in/m_vidyasagar/AI-7100/Aug-2024/
- This URL contains:
 - Course notes as of January 2025
 - Slides of lectures
- The citations below are in the bibliography of the course notes.
- The book will be published by Springer-Verlag, London.

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Invex Functions

What is so special about *convex* functions?

⁴References are found in the incomplete Class Notes.

Invex Functions

What is so special about *convex* functions? Every stationary point is also a *global minimum*, because

$$J(\phi) \geq J(\theta) + \langle \nabla J(\theta), \phi - \theta \rangle.$$

So $\nabla J(\theta^*) = \mathbf{0}$ implies that $J(\phi) \geq J(\theta^*)$ for all ϕ .

Do *other classes* of functions have this property?

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Invex Functions

What is so special about *convex* functions? Every stationary point is also a *global minimum*, because

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So $\nabla J(\theta^*) = \mathbf{0}$ implies that $J(\phi) \geq J(\theta^*)$ for all ϕ .

Do *other classes* of functions have this property?

Hanson (1981)⁴ introduced *invex* functions which need not be convex, but have this property.

Craven-Glover (1985) proved the converse!

Cute result, but not directly useful.

⁴References are found in the incomplete Class Notes.

Functions of Class PL

Lemma

(See JOTA paper, Lemma 4.1) Suppose $\nabla J(\cdot)$ is L -Lipschitz-continuous, and that $J^* = 0$. Then

$$\|\nabla J(\boldsymbol{\theta})\|_2^2 \leq 2LJ(\boldsymbol{\theta}).$$

Definition

A $\mathcal{C}^1$ function $J(\cdot)$ belongs to the class **Polyak-Lojasiewicz (PL)** if there exists a constant K such that

$$\|\nabla J(\boldsymbol{\theta})\|_2^2 \geq KJ(\boldsymbol{\theta}), \forall \boldsymbol{\theta} \in \mathbb{R}^d.$$



Example of a Nonconvex Function of Class PL

Define $J : \mathbb{R} \rightarrow \mathbb{R}$ via

$$J(\theta) = \theta^2 + 3 \sin^2 \theta.$$

A plot of $J(\cdot)$ is shown on the next slide.

Clearly $J(\cdot)$ is not convex, but it satisfies the PL condition with $K \approx 0.3511$.

Graph of a Function of Class PL

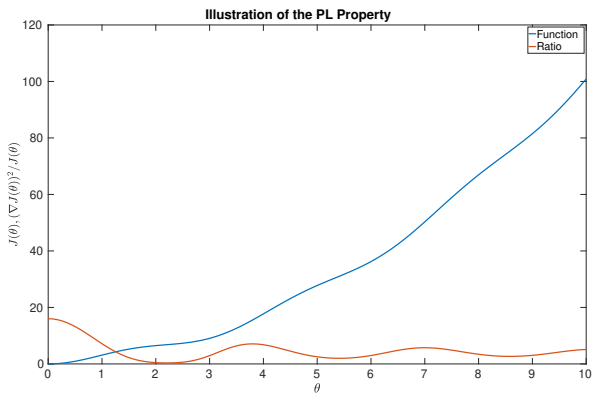


Figure: Graphs of $J(\theta) = \theta^2 + 3 \sin^2 \theta$, and of $|J'(\theta)|^2 / J(\theta)$.

Functions of Class $\mathcal{B}$

Definition

A function $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is said to belong to **Class $\mathcal{B}$** if $\phi(0) = 0$, and in addition, for arbitrary real numbers $0 < \epsilon \leq M$, it is true that

$$\inf_{\epsilon \leq r \leq M} \phi(r) > 0.$$

Example

Define a function $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ by

$$\phi(\theta) = \begin{cases} \theta, & \text{if } \theta \in [0, 1], \\ e^{-(\theta-1)}, & \text{if } \theta > 1. \end{cases}$$

Then ϕ belongs to Class $\mathcal{B}$.

Graph of a Function of Class $\mathcal{B}$

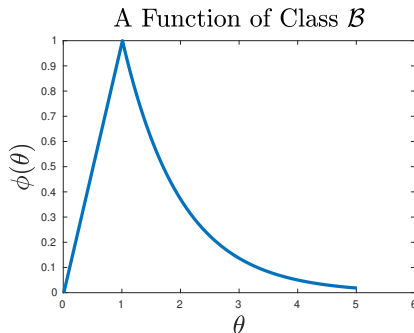


Figure: An illustration of a function in Class $\mathcal{B}$

Functions of Class KL'

Definition

A function $J(\cdot)$ is said to belong to the modified class **Kurdyka-Lojasiewicz (KL')** if there exists a function $\psi(\cdot)$ of Class $\mathcal{B}$ such that

$$\|\nabla J(\boldsymbol{\theta})\|_2 \geq \psi(J(\boldsymbol{\theta})), \forall \boldsymbol{\theta} \in \mathbb{R}^d.$$

Example of a Function of Class KL'

Consider an even function $J : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$J^*(\theta) = \begin{cases} \theta^2 + 4 \sin^2 \theta, & 0 \leq \theta \leq 5, \\ J(5) + 0.5J'(5)(1 - \exp(-2(\theta - 5))), & \theta > 5, \\ J(-\theta), & \theta < 0. \end{cases}$$

Then $J(\cdot)$ belongs to Class KL'.

Plot of a Function of Class KL'

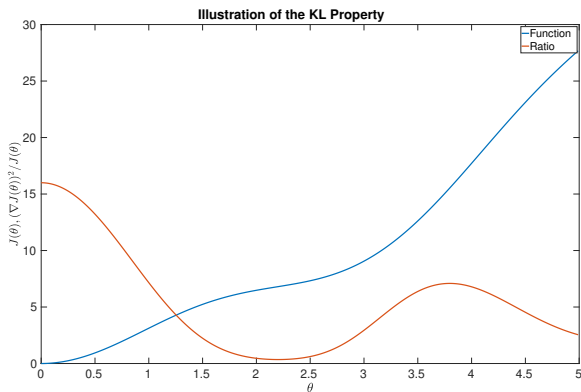


Figure: Graph of a function of Class KL'

Significance of the Classes PL and KL'

- PL implies KL' which implies invexity. Thus every stationary point is a global minimum.
- Both classes contain nonconvex functions.
- For functions of Class (KL'), we establish the convergence of various algorithms, but *without* any rates.
- For functions of Class (PL), we not only establish the convergence of various algorithms, but also derive estimates on the *rate* of convergence.

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Filtrations

All random variables are defined on an underlying probability space (Ω, Σ, P) where Ω is some set, Σ is a σ -algebra of subsets of Ω , and P is a probability measure on (Ω, Σ) .

A **filtration** $\{\mathcal{F}_t\}$ is an increasing sequence of σ -algebras in $\mathcal{F}$.
Thus

$$\mathcal{F}_t \subseteq \mathcal{F}_{t+1} \subseteq \mathcal{F}.$$

If $X_1, \dots, X_m$ are random variables on (Ω, Σ, P) , the σ -algebra generated by them is the smallest σ -algebra $\Sigma(X_1, \dots, X_m)$ such that each X_i is measurable with respect to $(\Omega, \Sigma(X_1, \dots, X_m))$.



Conditional Expectations

If X is an $\mathbb{R}^d$ -valued r.v., and $\mathcal{F}_t$ is some sub-algebra of $\mathcal{F}_t$, the **conditional expectation** of X w.r.t. $\mathcal{F}_t$ is the best approximation to X by a function measurable w.r.t. $(\Omega, \mathcal{F}_t)$, and is denoted by $E(X|\mathcal{F}_t)$. If $\{\mathcal{F}_t\}$ is a filtration, we denote $E(X|\mathcal{F}_t)$ by $E_t(X)$.

The **conditional variance** of X is defined as

$$CV_t(X) = E_t(\|X - E_t(X)\|_2^2) = E_t(\|X\|_2^2) - (E_t(X))^2.$$

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"Unified" Algorithm

Given C^1 -function $J : \mathbb{R}^d \rightarrow \mathbb{R}$.

Objective: Generate a (random) sequence $\{\theta_t\}$ in $\mathbb{R}^d$, which we hope converges to a global minimizer θ^* of $J(\cdot)$.

The iterative algorithm is defined in terms of two auxiliary vectors, denoted by $\mathbf{w}_t$ and $\mathbf{v}_t$, such that

$$\theta_t = \mathbf{w}_t - \epsilon_t \mathbf{v}_t, \mathbf{w}_t = \theta_t + \epsilon_t \mathbf{v}_t.$$

The updating formulas for $\mathbf{w}_t$ and $\mathbf{v}_t$ are as follows:

$$\mathbf{w}_{t+1} = \mathbf{w}_t + a_t \mathbf{v}_t - b_t \alpha_t \mathbf{h}_{t+1},$$

$$\mathbf{v}_{t+1} = \mu_t \mathbf{v}_t - \alpha_t \mathbf{h}_{t+1},$$

where $\mathbf{h}_{t+1}$ is a *stochastic gradient*, i.e., a random approximation to $\nabla J(\mathbf{w}_t)$ (*not* $\nabla J(\theta_t)$).

Special Case: Stochastic Gradient Descent

Several popular algorithms are special cases of the above, with suitable choices of parameters. In all cases, α_t is called the “step size,” or “learning rate,” while μ_t is called the “momentum parameter.”

Choose $\epsilon_t = 0$, $a_t = 0$, $b_t = 1$, $\mu_t = 0$, and α_t as desired. This gives the **Stochastic Gradient Descent**, wherein $\mathbf{w}_t = \boldsymbol{\theta}_t$, and

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \alpha_t \mathbf{h}_{t+1},$$

where $\mathbf{h}_{t+1}$ is a random approximation to $\nabla J(\boldsymbol{\theta}_t)$.

Special Case: Stochastic Heavy Ball

Choose $\epsilon_t = 0$, $a_t = \mu_t$, $b_t = 1$, and α_t, μ_t as desired. This gives the **Stochastic Heavy Ball** method, wherein

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \mu_t(\boldsymbol{\theta}_t - \boldsymbol{\theta}_{t-1}) - \alpha_t \mathbf{h}_{t+1},$$

and $\mathbf{h}_{t+1}$ is a random approximation to $\nabla J(\boldsymbol{\theta}_t)$.

Special Case: Stochastic Nesterov Accelerated Gradient

Choose $\epsilon_t = 1 \forall t$, $a_t = \mu_{t+1}\mu_t$, $b_t = 1 + \mu_{t+1}$, and α_t, μ_t as desired. This gives the **Stochastic Nesterov Accelerated Gradient** method, wherein

$$\theta_{t+1} = \theta_t + \mu_t(\theta_t - \theta_{t-1}) - \alpha_t \mathbf{h}_{t+1}.$$

and $\mathbf{h}_{t+1}$ is a random approximation to $\nabla J(\theta_t + \mu_t(\theta_t - \theta_{t-1}))$, and not $\nabla J(\theta_t)$.

Some Definitions Re: the Stochastic Gradient

Recall: $\mathbf{h}_{t+1}$ is a random approximation to $\nabla J(\mathbf{w}_t)$. So define $\mathcal{F}_t$ to be the σ -algebra generated by $\theta_0^t, \mathbf{h}_1^t$. Further, define

$$\mathbf{z}_t = E_t(\mathbf{h}_{t+1}), \quad \mathbf{x}_t = \mathbf{z}_t - \nabla J(\mathbf{w}_t), \quad \boldsymbol{\zeta}_{t+1} = \mathbf{h}_{t+1} - \mathbf{z}_t.$$

Interpretation: $\mathbf{x}_t$ is the *bias* of the stochastic gradient. If $\mathbf{x}_t = \mathbf{0}$ for all t , then $E_t(\mathbf{h}_{t+1}) = \nabla J(\mathbf{w}_t)$.

Further, $\boldsymbol{\zeta}_{t+1}$ is the *unpredictable part* of $\mathbf{h}_{t+1}$, and $E_t(\|\boldsymbol{\zeta}_{t+1}\|_2^2)$ is the *conditional variance* of $\mathbf{h}_{t+1}$.

Assumptions Re: the Stochastic Gradient

Assumptions: There exist sequences of constants $\{\mu_t\}$ and $\{M_t\}$ such that

$$\|\mathbf{x}_t\|_2 \leq B_t[1 + \|\nabla J(\mathbf{w}_t)\|_2], \quad \forall \boldsymbol{\theta}_t \in \mathbb{R}^d, \quad \forall t,$$

$$E_t(\|\zeta_{t+1}\|_2^2) \leq M_t^2[1 + J(\mathbf{w}_t)], \quad \forall \boldsymbol{\theta}_t \in \mathbb{R}^d, \quad \forall t.$$

If the measurement error is *additive*, it is reasonable to assume that $B_t = 0$ and $M_t^2 \leq M^2$ for all t .

In general, $B_t \rightarrow 0$ as $t \rightarrow \infty$, while M_t can be unbounded, as in zeroth-order methods.

Assumptions on the Constants

- 1 There exist constants $\bar{a}, \underline{b}, \bar{b}, \bar{\mu}, \bar{\epsilon}$ such that, for all t , we have

$$0 \leq a_t \leq \bar{a}, 0 < \underline{b} \leq b_t \leq \bar{b}, 0 \leq \bar{\mu} < 1, |\epsilon_t| \leq \bar{\epsilon} < \infty.$$

- 2 Define

$$k_t := \frac{a_t}{(1 - \mu_t)}, \bar{k} := \frac{\bar{a}}{1 - \bar{\mu}},$$

Then

$$k_t - k_{t+1} \rightarrow 0, \text{ as } t \rightarrow \infty.$$

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Theorem 1

Theorem

① Suppose

$$\sum_{t=0}^{\infty} \alpha_t^2 < \infty, \quad \sum_{t=0}^{\infty} \alpha_t B_t < \infty, \quad \sum_{t=0}^{\infty} \alpha_t^2 M_t^2 < \infty.$$

Then $\{\nabla J(\boldsymbol{\theta}_t)\}$ and $\{J(\boldsymbol{\theta}_t)\}$ are bounded a.s.

② If in addition $J(\cdot)$ satisfies (KL'), and

$$\sum_{t=0}^{\infty} \alpha_t = \infty,$$

then $J(\boldsymbol{\theta}_t) \rightarrow 0$ and $\nabla J(\boldsymbol{\theta}_t) \rightarrow \mathbf{0}$ as $t \rightarrow \infty$ a.s.

Corollary 1

Corollary

(Useful for additive noise) Suppose $B_t = 0$ and $M_t^2 \leq M^2$ for all t .

① Suppose

$$\sum_{t=0}^{\infty} \alpha_t^2 < \infty.$$

Then $\{\nabla J(\boldsymbol{\theta}_t)\}$ and $\{J(\boldsymbol{\theta}_t)\}$ are bounded a.s.

② If in addition $J(\cdot)$ satisfies (KL'), and

$$\sum_{t=0}^{\infty} \alpha_t = \infty,$$

then $J(\boldsymbol{\theta}_t) \rightarrow 0$ and $\nabla J(\boldsymbol{\theta}_t) \rightarrow \mathbf{0}$ as $t \rightarrow \infty$ a.s.

Interpretation of Corollary 1

- The conditions for convergence are the well-known **Robbins-Monro** conditions from 1951.
- Till date no one had succeeded in proving convergence under these general conditions.

Corollary 2

Corollary

(Useful for zeroth-order methods.) Suppose $B_t = O(c_t)$, and $M_t^2 = O(1/c_t^2)$.

① *Suppose*

$$\sum_{t=0}^{\infty} \alpha_t^2 < \infty, \quad \sum_{t=0}^{\infty} \alpha_t c_t < \infty, \quad \sum_{t=0}^{\infty} (\alpha_t^2)/(c_t^2) < \infty.$$

Then $\{\nabla J(\theta_t)\}$ and $\{J(\theta_t)\}$ are bounded a.s.

② *If in addition $J(\cdot)$ satisfies (KL'), and*

$$\sum_{t=0}^{\infty} \alpha_t = \infty,$$

then $J(\theta_t) \rightarrow 0$ and $\nabla J(\theta_t) \rightarrow \mathbf{0}$ as $t \rightarrow \infty$ a.s.

Interpretation of Corollary 2

- The conditions for convergence are the well-known **Kiefer-Wolfowitz-Blum** conditions from 1952 and 1954 when zeroth-order methods are used.
- Till date no one had succeeded in proving convergence under these general conditions.

Theorem 2

If we strengthen the assumption on $J(\cdot)$ from (KL') to (PL), and keep the other assumptions the same, then we get bounds in the *rate* of convergence of $J(\theta_t)$ to zero, for *any* sequences $\{B_t\}$ and $\{M_t\}$ that satisfy the hypotheses.

Not stated here, but are in the PAFA paper and the arxiv version.

Near Optimality of the Rate of Convergence

Arjevani et al. (2023) have shown that, if $B_t = 0$ (unbiased stochastic gradient) and $M_t^2 \leq M^2$ for all t (uniformly bounded conditional variance), and if $J(\cdot)$ is convex, then $J(\theta_t) = O(t^{-1})$ in the worst case, for *any* algorithm.

We shown that, in this case, for functions of Class (PL), $J(\theta_t) = o(t^{-\lambda})$ for all $\lambda \in (0, 1)$.

In short, the convergence rate of our algorithm is *nearly optimal, and applies to some nonconvex functions as well.*

Alternate Forms of Updating

Because our proofs make use of martingale convergence theory, we can prove that the unified algorithm converges under the stated hypotheses for

- Minibatch as opposed to batch
- Zeroth-order gradient approximation
- Block updating of iterations
- Any combination of the above.

Outline

1 The "Why" of Stochastic Gradient Descent

- Background
- Origins and Benefits of Stochasticity

2 Problem Formulations

- Classes of Objective Functions
- Quick Primer on Stochastic Processes

3 Convergence Theorems

- Algorithms Studied
- Convergence Theorems for Unified Algorithm
- Numerical Results and Future Research Topics

Objective Function

We minimized an objective function with $d = 10^6$ of the form

$$J(\boldsymbol{\theta}) = \frac{1}{2} \boldsymbol{\theta}^\top A \boldsymbol{\theta} + c \sin^2(\|\boldsymbol{\theta}\|_2^2),$$

where A was a 1000×1000 block Hilbert matrix, where each block was itself a 1000×1000 Hilbert matrix. Thus it is *very* badly conditioned. The constant c is chosen to ensure that $J(\cdot)$ belongs to Class PL.

Brief Summary of Numerical Results

- We simulated not only SGD, SHB and SNAG, but also ADAM, Adagrad, and RMSprop.
- With just additive Gaussian noise in the gradient, all methods besides SGD performed more or less equally well.
- However, when we used “zeroth-order” approximations (with biased gradients and increasing variance), SNAG *diverged*. The rest converged with SHB being a clear winner.

Scope for Future Research: Near Areas

- Our results are quite comprehensive, with *one exception*.
- Our results require that the momentum term μ_t satisfies $0 \leq \mu_t \leq \bar{\mu} < 1$.
- This is true for the standard version of Stochastic Heavy Ball, but *not true* for the standard version of Stochastic Nesterov Accelerated Gradient method.
- The “optimal” schedule for μ_t in SNAG is $\mu_t = (t + 2)/(t + 5)$; see Su, Boyd and Candès (2016).
- There is still no proper convergence theory for this schedule for μ_t , and for nonconvex objective functions.

Scope for Future Research (Cont'd)

- Are there other classes of invex functions that are readily identifiable, besides PL and KL?
- There are other methods that are widely used such as ADAM, RMSprop etc., to which our analysis doesn't apply. As of now there are no proofs of convergence for many of these algorithms, only simulations.

Scope for Future Research: Farther Area

- Please see: Liu Ziyin, Yuanjie Ren, Adam Levine, and Isaac Chuang, "Thermodynamic Irreversibility of Training Algorithms," arxiv:2605.21933
- This article presents an alternate approach to the analysis of SGD, using ideas from statistical physics. This work complements my approach based on martingale theory.
- Also see; Liu Ziyin, Mingze Wang, Hongchao Li and Lei Wu, "Parameter Symmetry and Noise Equilibrium of Stochastic Gradient Descent," arxiv: 2402.07193v3 This paper exploits the *symmetries* that are inherent in many optimization problems, eg., neural networks.

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Thank You!

